POLSCI 9592

Lecture 3: Interactions

Dave Armstrong

Goals for This Session

1. Discuss Interaction Effects in the Linear Model
2. Discuss Interaction Effects in the GLM
3. Presentation of Interaction Results

Interaction Effects

When the partial effect of one variable depends on the value of another variable, those two variables are said to "interact". Alternatively, we say that the relationship is *conditional*.

- For example, we may want to test whether age effects are different for men (coded 1) and women (coded 0).
- In such cases we could fit separate regressions for men and women, but this does not allow for a formal statistical test of the differences and only really works as intended in the linear model.
- Specification of interaction effects facilitates statistical tests for a difference in slopes within a single regression

Unconditional Effects in the LM

Consider the following linear model:

$$y_i = b_0 + b_1 x_i + b_2 z_i + e_i$$

In this model, the effect of x is:

$$\frac{\partial y}{\partial x} = b_1$$

The marginal effect is constant.

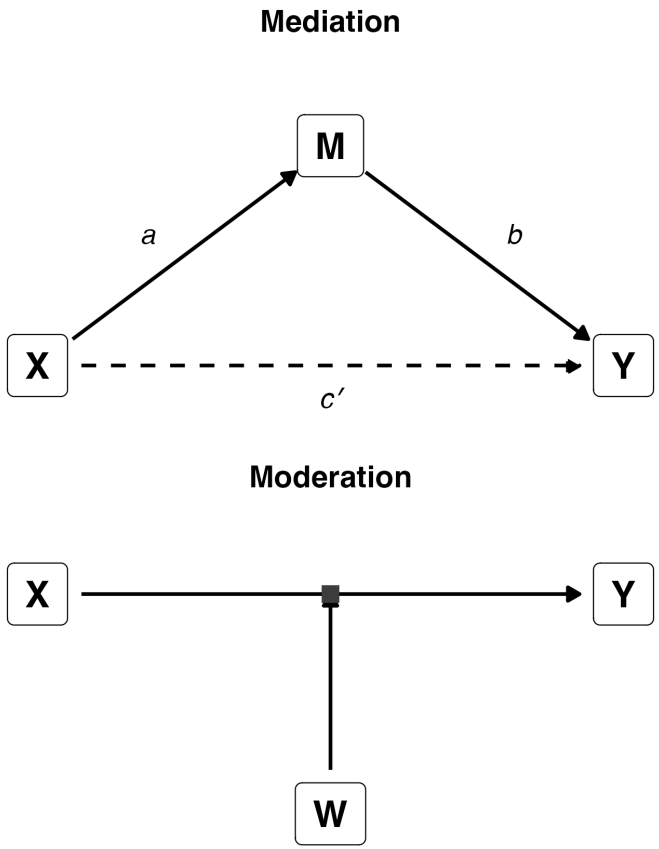
Mediation vs Moderation

Moderation asks - does the direct effect of X vary by some other variable, W?

- This is the effect heterogeneity of interest today.

Mediation asks - is the effect of X direct and/or indirect through some other variable, M?

- This is a related, but quite different question that we will not discuss today.



Moderation Effect

Suppose we are estimating a relationship between y and x and we suspect/hypothesize that relationship is moderated by z.

- We are really saying:

$$\frac{\partial y}{\partial x} = f(z)$$

The simplest $f(z)$ would just be linear:

$$\frac{\partial y}{\partial x} = b_1 + b_2 z_i$$

What model would generate this marginal effect:

$$y_i = b_0 + b_1 x_i + b_2 x_i z_i + b_3 z_i + \mathbf{gw}_i + e_i$$

The marginal effect of x from the above equation is $b_1 + b_2 z_i$.

Typology of Moderation Models

We can fit different kinds of moderation models depending on our hypotheses:

Typology of Models

Model	Assumption Change
Single interaction	Single effect varies across groups
Full interaction	All effects vary across groups
Subgroup Analyses	All effects + residual variances vary across groups

Interaction Effects

We often denote product regressors $x_i z_i$ as *interaction effects*.

Ultimately we want to know three things:

- Is there statistical evidence of a moderated effect?
- How variable is the moderated effect?
- Does the variation change or qualify the substantive conclusion?

Below, we walk through all the possible two-way interaction scenarios and we will discuss how to answer these questions.

Two Categorical Variables

With two categorical variables, essentially you are estimating a different conditional mean for every pair of values across the two categorical variables. You could do that as follows:

```
load("../lecture2/ces0421.rda")
ces21 <- ces0421 |> filter(year == 2021)

mod <- lm(leader_lib ~ educ * agegrp, data=ces21
```

```
library(broom)
tidy(mod)
```

```
## # A tibble: 9 × 5
##   term                           estimate std.error statistic  p.value
##   <chr>                             <dbl>     <dbl>     <dbl>    <dbl>
## 1 (Intercept)                       35.2       5.16     6.83   9.08e-12
## 2 educHS/College                    10.3       5.29     1.95   5.15e- 2
## 3 educCollege Grad                  16.6       5.21     3.19   1.45e- 3
## 4 agegrp35-54                       -0.609     6.13    -0.0994 9.21e- 1
## 5 agegrp55+                          1.79      5.64     0.318  7.51e- 1
## 6 educHS/College:agegrp35-54        -4.45      6.32    -0.704  4.81e- 1
## 7 educCollege Grad:agegrp35-54      -4.45      6.21    -0.717  4.74e- 1
## 8 educHS/College:agegrp55+          -6.19      5.80    -1.07   2.85e- 1
## 9 educCollege Grad:agegrp55+       -10.8       5.71    -1.89   5.90e- 2
```

Evidence of Moderation

We evaluate evidence of moderation by evaluating the moderated model agains the unconditional model:

```
uncond_mod <- lm(leader_lib ~ educ + agegrp, data=ces21)
```

In this case, there is evidence of a moderated effect - `educ`'s effect is moderated by `agegrp` (and vice-versa).

```
anova(uncond_mod, mod, test="F")
```

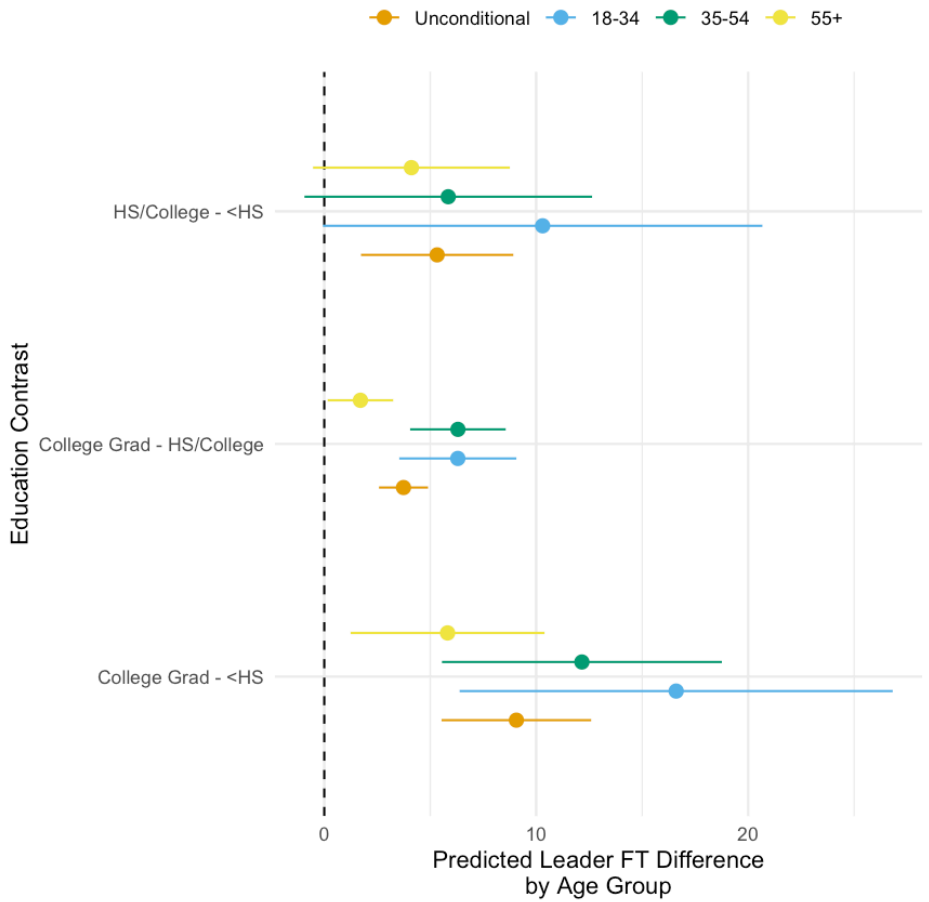
```
## Analysis of Variance Table
##
## Model 1: leader_lib ~ educ + agegrp
## Model 2: leader_lib ~ educ * agegrp
##   Res.Df      RSS Df Sum of Sq      F   Pr(>F)
## 1  14784 16130108
## 2  14780 16110130  4     19978 4.5821 0.001069 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The `anova()` function works because the unconditional model is nested in the conditional model.

Variation in Effects

There is considerable variation around the unconditional effect.

- The main conclusions appear to not hold up.



Two Categorical Summary

With these two categorical variables, the unconditional relationship fails to account for substantial heterogeneity across age groups.

- Especially the difference between HS/College and Univ is heterogeneous
  - Small for oldest group, larger for the other two.

One Categorical + One Numeric

With one categorical variable, d (we'll use a binary variable here) and one numeric variable, x, we would have

$$y_i = b_0 + b_1 x_i + b_2 d_i + b_3 x_i d_i + e_i$$

We can think of the moderation effect in two ways:

Marginal Effect of x

$$\frac{\partial y}{\partial x} = b_1 + b_3 d$$

Marginal effect of d

$$\frac{\partial y}{\partial d} = b_2 + b_3 x$$

Interaction of Education and Sex

```
data(SLID, package="carData")
mod2_0 <- lm(log(wages) ~ education + sex, data=SLID)
mod2_1 <- lm(log(wages) ~ education * sex, data=SLID)
tidy(mod2_1)
```

```
## # A tibble: 4 × 5
##   term              estimate std.error statistic   p.value
##   <chr>                <dbl>     <dbl>     <dbl>     <dbl>
## 1 (Intercept)         1.73     0.0492      35.3  1.69e-237
## 2 education           0.0577   0.00358     16.1  1.14e- 56
## 3 sexMale             0.477    0.0669       7.13 1.15e- 12
## 4 education:sexMale  -0.0186   0.00489     -3.81 1.41e-  4
```

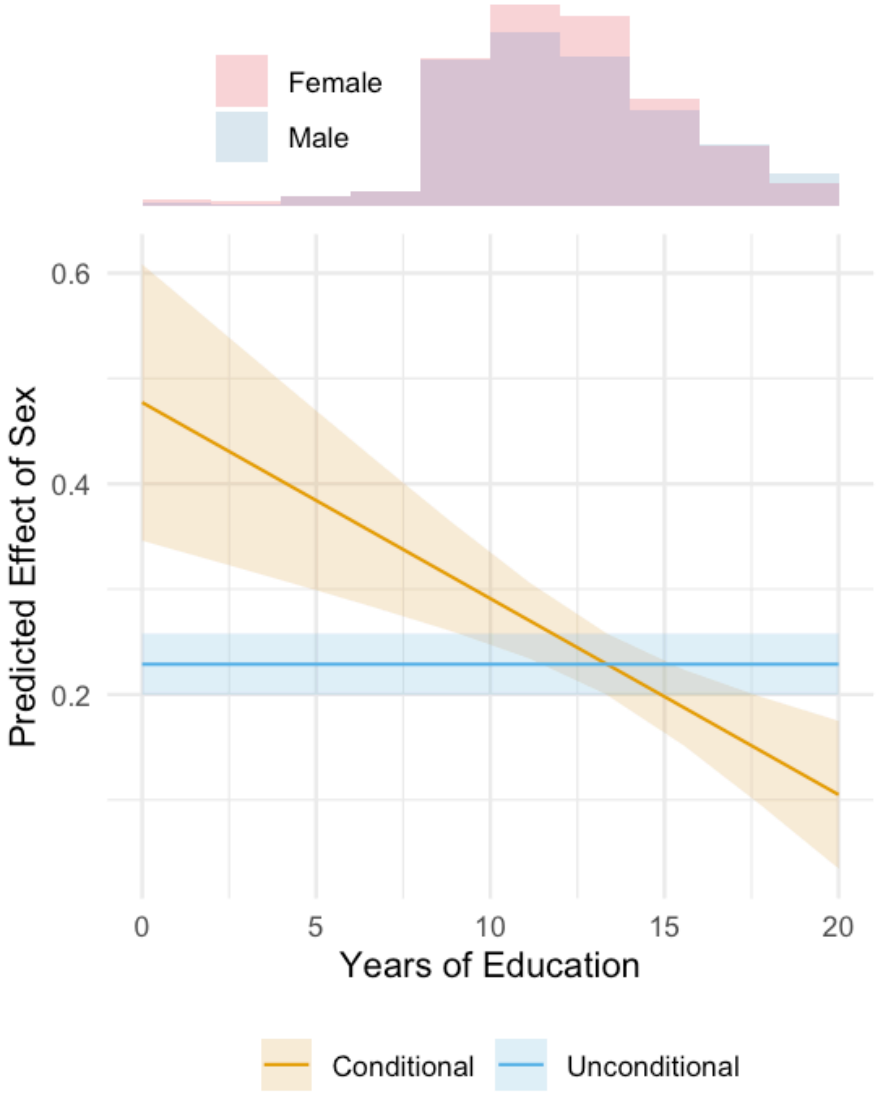
There is evidence of a moderation effect.

```
## # A tibble: 5 × 5
##   model                     sex                estimate std.error  p.value
##   <chr>                     <chr>                 <dbl>     <dbl>    <dbl>
## 1 Unconditional             Female               0.0477   0.00244 4.67e-85
## 2 Unconditional             Male                 0.0477   0.00244 4.67e-85
## 3 Conditional               Female               0.0577   0.00358 1.96e-58
## 4 Conditional               Male                 0.0391   0.00333 7.25e-32
## 5 Difference in Conditional (Male) - (Female)  -0.0186   0.00489 1.39e- 4
```

Effect of Sex

The unconditional effect is (necessarily) flat and the conditional effect is negatively sloped.

- Note the density of education - it marks the regions where comparisons are permissible.
- The effects are always positive (males are predicted to earn more than females regardless of education).
- The difference between salaries of females and males decreases with education.



Two Numeric Variables

With two numeric variables the interpretation gets a bit trickier. For example, consider the following model:

$$\hat{Y}_i = b_0 + b_1 X_{i1} + b_2 X_{i2} + b_3 X_{i3} + b_4 X_{i1} X_{i2}$$

We want to know the partial conditional effect of both X_1 and X_2, but unlike above, neither can be boiled down to a small set of values. Just think about the equation:

$$\frac{\partial \hat{Y}}{\partial X_1} = \beta_1 + \beta_4 X_2$$

$$\frac{\partial \hat{Y}}{\partial X_2} = \beta_2 + \beta_4 X_1$$

Note, that β_4 is the amount by which the *effect* of X_1 goes up for every additional unit of X_2 and the amount by which the *effect* of X_2 goes up for every additional unit of X_1.

Testable Hypotheses

$$\hat{Y}_i = b_0 + b_1 X_{i1} + b_2 X_{i2} + b_3 X_{i3} + b_4 X_{i1} X_{i2}$$

Berry, Golder and Milton (2012) suggest that we should be able to test 5 hypotheses:

- $\mathbf{P}_{X_1|X_2=\min}$ The marginal effect of X_1 is [positive, zero, negative] when X_2 takes its lowest value.
- $\mathbf{P}_{X_1|X_2=\max}$ The marginal effect of X_1 is [positive, zero, negative] when X_2 takes its highest value.
- $\mathbf{P}_{X_2|X_1=\min}$ The marginal effect of X_2 is [positive, zero, negative] when X_1 takes its lowest value.
- $\mathbf{P}_{X_2|X_1=\max}$ The marginal effect of X_2 is [positive, zero, negative] when X_1 takes its highest value.
- $\mathbf{P}_{X_1 X_2}$ The marginal effect of each of X_1 and X_2 is [positively, negatively] related to the other variable.

Example

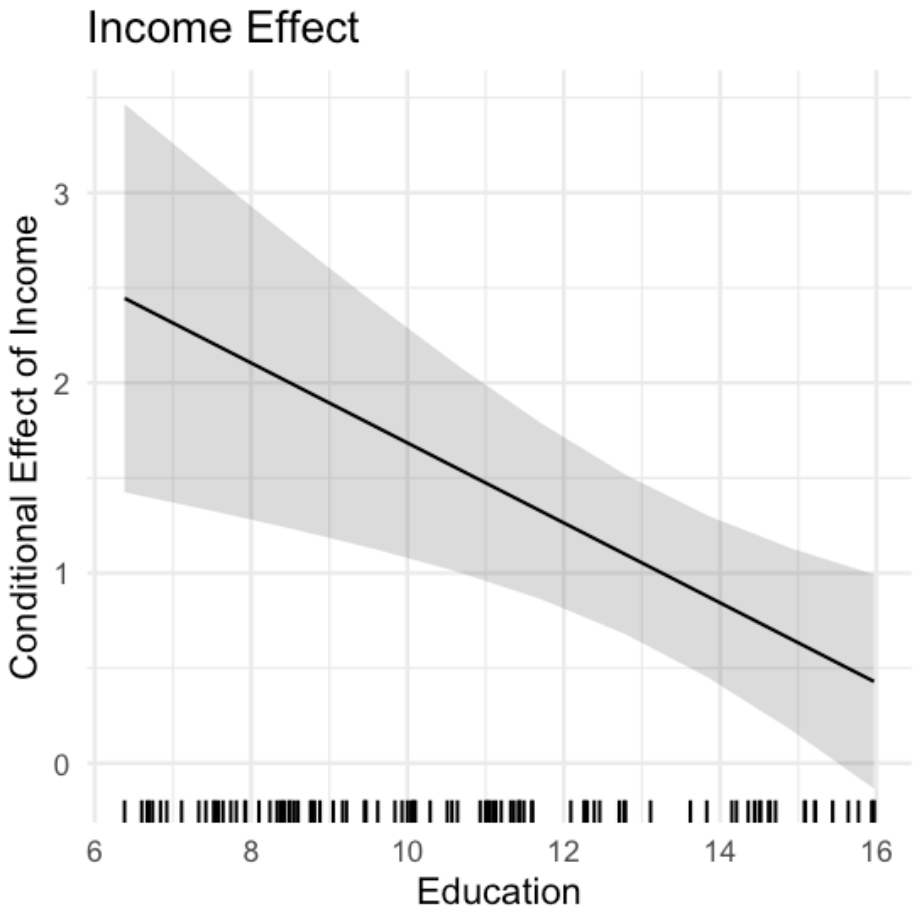
```
data(Prestige, package="carData")
Prestige$income <- Prestige$income/1000
addmod <- lm(prestige ~ income + education + type, data=Prestige)
intmod <- lm(prestige ~ income*education + type, data=Prestige)
tidy(intmod)
```

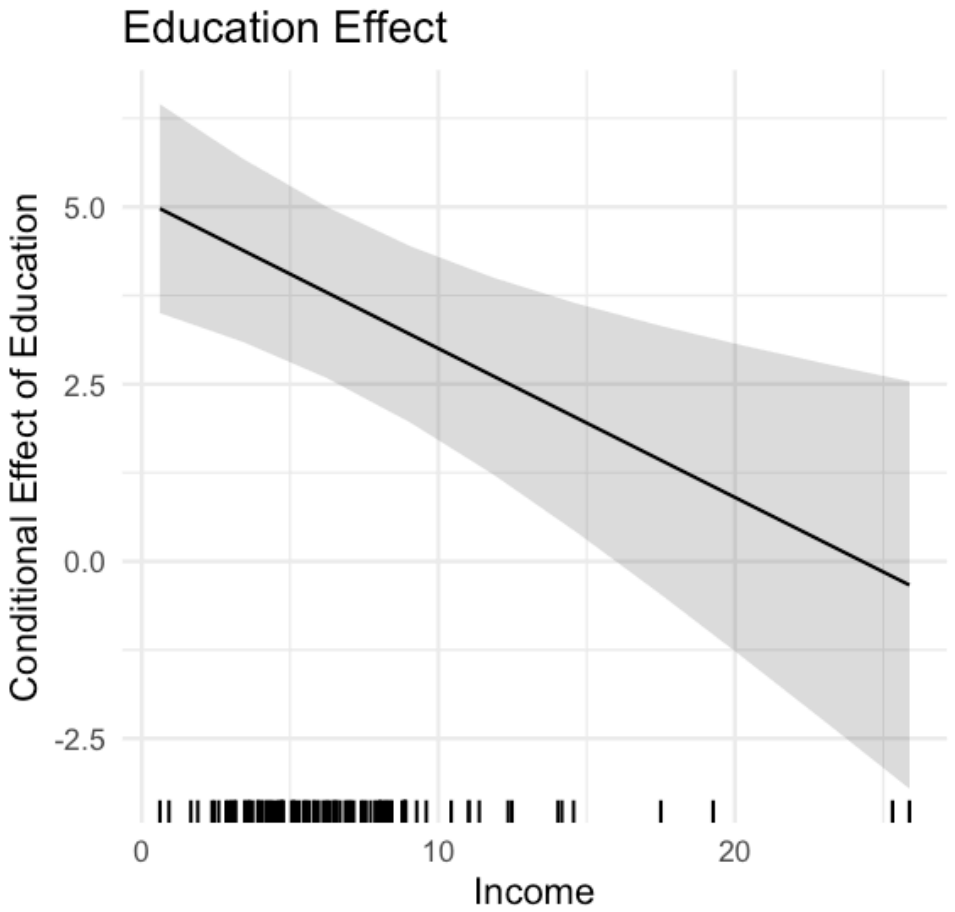
```
## # A tibble: 6 × 5
##   term             estimate std.error statistic       p.value
##   <chr>               <dbl>     <dbl>     <dbl>         <dbl>
## 1 (Intercept)       -17.8      7.59       -2.34 0.0212
## 2 income              3.79     0.945       4.01 0.000124
## 3 education           5.10     0.777       6.57 0.00000000293
## 4 typeprof            5.48     3.71        1.48 0.144
## 5 typewc             -3.58     2.43       -1.48 0.143
## 6 income:education   -0.210    0.0698     -3.01 0.00335
```

Evidence of Moderation

- The `income:education` line answers this question. If it is significant, then there is a significant interaction, otherwise there is not.
- This is counter to a minor, though still influential, point in Brambor, Clark and Golder (2006), but is consistent with Berry, Golder and Milton (2012).
- The interaction provides evidence of moderation.

Conditional Effects





Interpretation

- The effect of income is nearly always significant, though it gets smaller as education gets bigger. That is, as education increases, we expect smaller increases in prestige from increasing income
- The effect of education is significant and positive until around 16,000 dollars, which is around 2/3 the range of `income`, but is the 96th percentile because of the skewness of income.
- This suggests that people tend to derive prestige from either higher incomes or higher education, but not really both.

Berry, Golder and Milton Test

```
BGMtest(intmod, vars = c("income", "education"))
```

```
##              est    se      t p-value
## P(X|Zmin)  2.445 0.520  4.698 0.000
## P(X|Zmax)  0.429 0.287  1.495 0.138
## P(Z|Xmin)  4.756 0.712  6.681 0.000
## P(Z|Xmax) -0.335 1.466 -0.229 0.820
## P(XZ)     -0.210 0.070 -3.012 0.003
```

When Confidence Bounds Equal Zero

You may want to know when the confidence bounds are equal to zero. Consider the equation:

$$\hat{Y}_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \beta_3 X_{i3} + \beta_4 X_{i1} X_{i2}$$

- We know that the conditional effect of X_1 is $\beta_1 + \beta_4 X_2$ and that the lower bound is $(\beta_1 + \beta_4 X_2) - 1.96 \times SE(\beta_1 + \beta_4 X_2)$.
- Since those are all quantities that we know (or estimate), we could set it equal to zero and solve.
- This is what the `changeSig` function does.

Change in Significance

```
changeSig(intmod, c("income", "education"))
```

```
## LB for B(income | education) = 0 when education=15.4979 (95th pctile)
## UB for B(income | education) = 0 when education=27.9396 (> Maximum Value in Data)
## LB for B(education | income) = 0 when income=15.9273 (96th pctile)
## UB for B(education | income) = 0 when income=59.5175 (> Maximum Value in Data)
```

Transition from Linear to GLM

Consider a linear model: $y = b_0 + b_1 x + b_2 z + e$

- The effect of x is $\frac{\partial y}{\partial x} = b_1$

In the non-linear model: $E(y|x, z) = g^{-1}(b_0 + b_1 x + b_2 z) = g^{-1}(\eta)$

- The effect of x is

$$\frac{\partial g^{-1}(\eta)}{\partial x} = \frac{\partial g^{-1}(\eta)}{\partial \eta} \frac{\partial \eta}{\partial x}$$

Common Links and First Derivatives

Model	Response Function	Effect of x
Linear	$E(Y) = \eta$	β_x
Binary	$p = F(\eta)$	$\beta_x f(\eta)$

where

- $F(\cdot)$ is a cumulative distribution function (CDF)
- $f(\cdot) = F'(\cdot)$ is the corresponding probability density function (PDF)

Binary Model Example

Consider the binary regression model:

$$\log\left(\frac{p}{1-p}\right) = b_0 + b_1 x + b_2 z$$

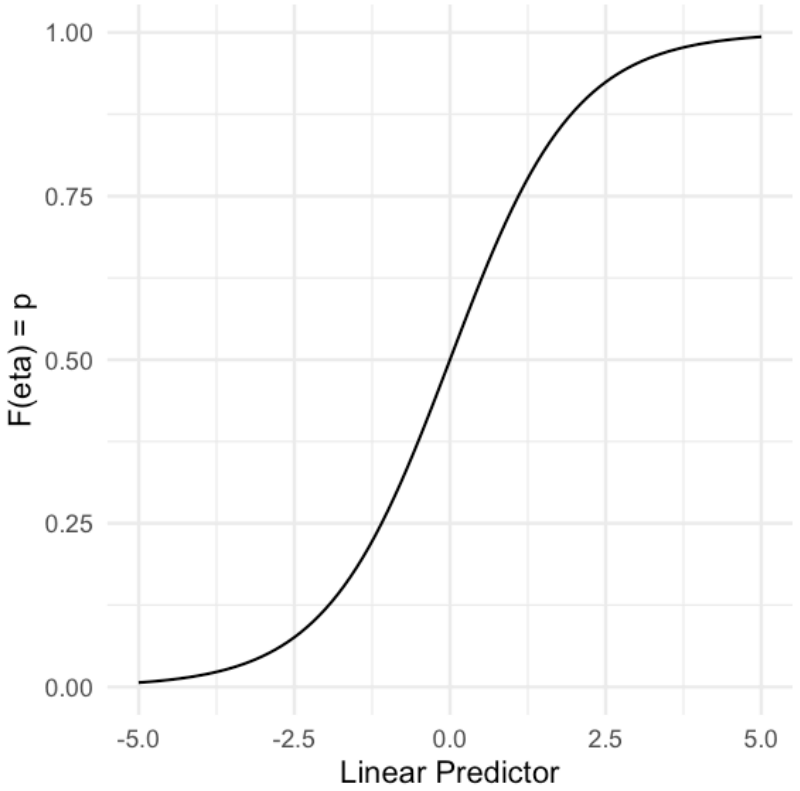
The effect of x is:

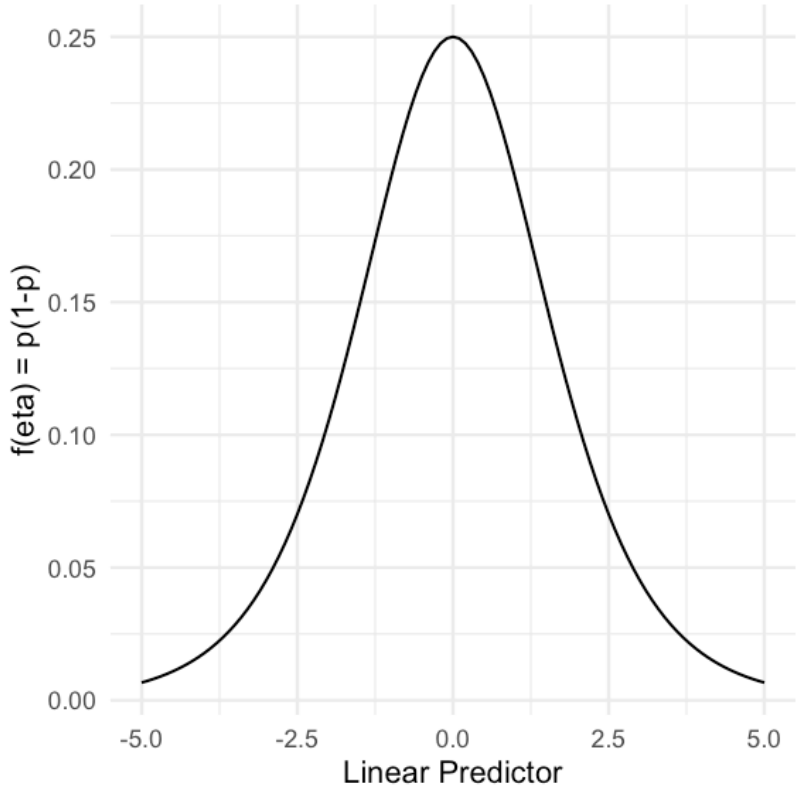
$$\frac{\partial E(y|x,z)}{\partial x} = f(\eta) b_1 = p(1-p) b_1$$

Because p depends on all predictors, the effect of x changes across values of x and z.

Effect Visualization

The effect is biggest when $p = 0.5$ and gets smaller in either direction.





Link vs Product Term Conditionality

Mechanical conditionality that comes from the non-linear link:

- *can* dampen or amplify the coefficient of x as a function of z.
- *cannot* allow a marginal effect to change signs, but the second difference still could.

Including a product term:

- *can* permit sign flips based on different values of the conditioning variable.
- *can* reduce the level of conditionality between x and z in some circumstances.

Link vs Product Term Conditionality

If $E(y|x,z) = g^{-1}(b_0 + b_1 x + b_2 z)$

$$\frac{\partial g^{-1}(\eta)}{\partial x} = \frac{\partial g^{-1}(\eta)}{\partial \eta} b_1 = f(\eta) b_1$$

If $E(y|x,z) = g^{-1}(b_0 + b_1 x + b_2 z + b_3 xz)$

$$\frac{\partial g^{-1}(\eta)}{\partial x} = \frac{\partial g^{-1}(\eta)}{\partial \eta} (b_1 + b_3 z) = f(\eta)(b_1 + b_3 z)$$

Since $f(\eta)$ is a density, it is non-negative

Conditionality in GLMs

If the model induces conditionality by definition, how do we think about moderation?

1. Just as before - generally, an investigation about conditionality should be theoretically interesting.
    - At this stage, you should not go "hunting for" interactions.
2. Unlike before, the mechanical model building exercise does not lead to clear, immediate answers.
    - Whether the product term is significant and the extent of the moderation are less related here.
3. Whether a product term belongs in the model depends on whether it generates substantively important conditionality beyond the mechanical conditionality induced by the link function.

Evaluating Conditionality

Variable Types	Marginal Effect of x	Interaction Quantity
x, z both continuous	$\frac{\partial E(Y)}{\partial x}$	$\frac{\partial}{\partial z}\frac{\partial E(Y)}{\partial x}$
x continuous, z binary	$\frac{\partial E(Y)}{\partial x}$	$\Delta_z\left(\frac{\partial E(Y)}{\partial x}\right)$
x, z both binary	$\Delta_x E(Y)$	$\Delta_z\left(\Delta_x E(Y)\right)$

Second differences work for any variable types. They are required whenever a variable is categorical, and for continuous variables the step sizes you choose define the quantity.

The Cross-Partial in the Logit Model

For the logit, $\frac{\partial p}{\partial \eta} = p(1-p)$ and $\frac{\partial^2 p}{\partial \eta^2} = p(1-p)(1-2p)$.

Additive: $\eta = b_0 + b_1 x + b_2 z$

$$\frac{\partial^2 p}{\partial x \partial z} = b_1 b_2 \, p(1-p)(1-2p)$$

- Non-zero whenever b_1 and b_2 are, except at $p = 0.5$.
- Changes sign at $p = 0.5$.
- The marginal effect of x cannot change sign, but the interaction can.

Product: $\eta = b_0 + b_1 x + b_2 z + b_3 xz$

$$\frac{\partial^2 p}{\partial x \partial z} = b_3 \, p(1-p) + (b_1 + b_3 z)(b_2 + b_3 x) \, p(1-p)(1-2p)$$

- The second term is compression; it remains even if $b_3 = 0$.
- b_3 alone does not give the size, or even the sign, of the interaction (Ai and Norton 2003).

Example: ANES Voting Democrat

```
anes <- rio::import("anes1992.dta")
anesu <- anes |>
  mutate(retnat = rio::factorize(retnat),
         educ = rio::factorize(educ)) |>
  drop_na(votedem, retnat, pid, age, male,
          educ, black, south)
add_mod <- glm(votedem ~ retnat + pid + age + male + educ +
                 black + south, data=anesu, family=binomial)
tidy(add_mod)
```

```
## # A tibble: 9 × 5
##   term        estimate std.error statistic  p.value
##   <chr>          <dbl>     <dbl>     <dbl>    <dbl>
## 1 (Intercept)   0.512     0.673      0.760  4.47e- 1
## 2 retnatsame    1.33      0.594      2.24   2.50e- 2
## 3 retnatworse   1.65      0.570      2.89   3.89e- 3
## 4 pid          -0.916     0.0557   -16.4    1.06e-60
## 5 age           0.0131    0.00566    2.32   2.05e- 2
## 6 male         -0.292     0.180     -1.62   1.04e- 1
## 7 educ          0.251     0.104      2.42   1.57e- 2
## 8 black         2.15      0.477      4.50   6.64e- 6
## 9 south         0.0129    0.221      0.0584 9.53e- 1
```

Two Categorical Variables: `retnat, male`

```
econ_comps <- avg_comparisons(add_mod,
  datagrid(male = c(0,1),
           grid_type="counterfactual"),
  variables = list(retnat="pairwise"),
  by = c("male"))

econ_comps |> select(contrast, male, estimate, p.value)
```

```
##
##        Contrast Estimate Pr(>|z|)
##  same - better    0.1630  0.01889
##  worse - better   0.2025  0.00235
##  worse - same     0.0395  0.15592
##  same - better    0.1598  0.01673
##  worse - better   0.1992  0.00180
##  worse - same     0.0394  0.15528
```

```
econ_comps |> hypotheses(~pairwise  | contrast) |>
  select(contrast, hypothesis, estimate, p.value)
```

```
##
##        Contrast  Hypothesis  Estimate Pr(>|z|)
##  same - better  (b4) - (b1) -3.26e-03    0.364
##  worse - better (b5) - (b2) -3.31e-03    0.363
##  worse - same   (b6) - (b3) -5.02e-05    0.769
```

Example: ANES Voting Democrat

```
prod_mod <- glm(votedem ~ male*retnat + pid + age + black + educ
                south, data=anesu, family=binomial)
tidy(prod_mod)
```

```
## # A tibble: 11 × 5
##    term              estimate std.error statistic  p.value
##    <chr>                <dbl>     <dbl>     <dbl>    <dbl>
##  1 (Intercept)       -1.30     1.44      -0.899  3.69e- 1
##  2 male               2.04     1.54       1.33   1.84e- 1
##  3 retnatsame         3.02     1.44       2.10   3.56e- 2
##  4 retnatworse        3.49     1.41       2.47   1.34e- 2
##  5 pid               -0.916    0.0558   -16.4    1.21e-60
##  6 age                0.0130   0.00568    2.30   2.17e- 2
##  7 black              2.22     0.484      4.59   4.41e- 6
##  8 educ               0.259    0.104      2.48   1.31e- 2
##  9 south              0.0178   0.223      0.0801 9.36e- 1
## 10 male:retnatsame   -2.15     1.59      -1.36   1.75e- 1
## 11 male:retnatworse  -2.44     1.55      -1.57   1.16e- 1
```

```
lmtest::lrtest(add_mod, prod_mod)
```

```
## Likelihood ratio test
##
## Model 1: votedem ~ retnat + pid + age + male + educ + black + south
## Model 2: votedem ~ male * retnat + pid + age + black + educ + south
##   #Df  LogLik Df  Chisq Pr(>Chisq)
## 1   9 -401.05
```

Two Categorical Variables: `retnat, male`

```
econ_comps2 <- avg_comparisons(prod_mod,
  datagrid(male = c(0,1),
           grid_type="counterfactual"),
  variables = list(retnat="pairwise"),
  by = c("male"))

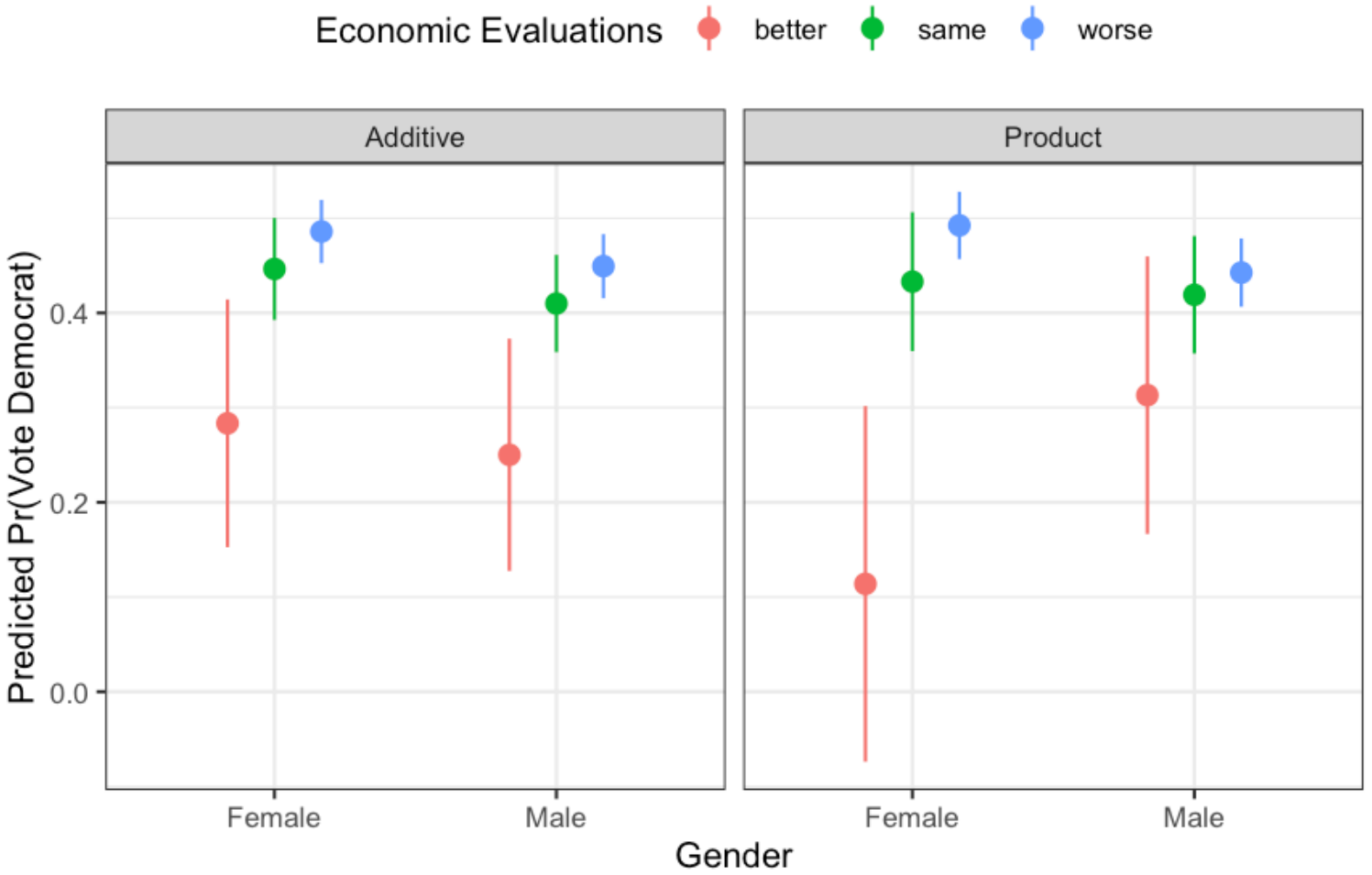
econ_comps2 |> select(contrast, male, estimate, p.value)
```

```
##
##         Contrast Estimate Pr(>|z|)
##  same - better    0.3192  0.00193
##  worse - better   0.3785  < 0.001
##  worse - same     0.0594  0.15491
##  same - better    0.1060  0.19121
##  worse - better   0.1295  0.09250
##  worse - same     0.0235  0.52272
```

```
econ_comps2 |> hypotheses(~pairwise  | contrast) |>
  select(contrast, hypothesis, estimate, p.value)
```

```
##
##         Contrast  Hypothesis Estimate Pr(>|z|)
##  same - better  (b4) - (b1)  -0.2131   0.1045
##  worse - better (b5) - (b2)  -0.2490   0.0452
##  worse - same   (b6) - (b3)  -0.0359   0.5157
```

Effects:


Economic Evaluations
better
same
worse
Additive
Product
Predicted Pr(Vote Democrat)
0.4
0.2
0.0
Female
Male
Female
Male
Gender

Example: ANES Voting Democrat: Additive

One Categorical, One Numeric: `pid, black`

```
slp2 <- avg_comparisons(add_mod,
    newdata = datagrid(black = c(0,1), pid = 1:6,
                       grid_type = "counterfactual"),
    variables = list(pid = 1),
    by = c("black", "pid"))
slp2 |> as_tibble() |>
  select(black, pid, estimate, p.value)
```

```
## # A tibble: 12 × 4
##    black   pid estimate  p.value
##    <dbl> <int>    <dbl>    <dbl>
##  1     0     1  -0.130  4.75e-78
##  2     0     2  -0.194  8.50e-79
##  3     0     3  -0.213  2.39e-63
##  4     0     4  -0.165  1.34e-80
##  5     0     5  -0.0950 1.27e-46
##  6     0     6  -0.0454 2.66e-17
##  7     1     1  -0.0242 2.10e- 2
##  8     1     2  -0.0538 9.38e- 3
##  9     1     3  -0.106  1.40e- 3
## 10     1     4  -0.174  8.36e- 7
## 11     1     5  -0.214  2.42e-50
## 12     1     6  -0.187  2.62e-10
```

```
slp2 |>
  hypotheses(~pairwise | pid) |>
  as_tibble() |>
  left_join(
    anesu |> count(pid, black) |>
      pivot_wider(names_from = "black", values_from = "n", names_
  select(pid, estimate, p.value, starts_with("black"))
```

```
## # A tibble: 6 × 5
##     pid estimate  p.value black_0 black_1
##   <dbl>    <dbl>    <dbl>   <int>   <int>
## 1     1  0.105   9.99e-17     151      48
## 2     2  0.140   1.10e- 9     142      24
## 3     3  0.106   1.50e- 3     140      11
## 4     4 -0.00965 7.78e- 1      71       6
## 5     5 -0.119   1.23e-11     131       2
## 6     6 -0.141   2.79e- 6     161       2
```

Example: ANES Voting Democrat

One Categorical, One Numeric: `pid, black`

```
slp2a <- avg_comparisons(prod_mod,
    newdata = datagrid(black = c(0,1), pid = 1:6,
                       grid_type = "counterfactual"),
    variables = list(pid = 1),
    by = c("black", "pid"))
slp2a |> as_tibble() |>
  select(black, pid, estimate, p.value)
```

```
## # A tibble: 12 × 4
##    black   pid estimate  p.value
##    <dbl> <int>    <dbl>    <dbl>
##  1     0     1  -0.130  1.21e-75
##  2     0     2  -0.196  1.14e-76
##  3     0     3  -0.216  2.66e-61
##  4     0     4  -0.166  1.36e-79
##  5     0     5  -0.0945 2.50e-44
##  6     0     6  -0.0444 3.68e-16
##  7     1     1  -0.0268 7.66e- 3
##  8     1     2  -0.0472 9.11e- 3
##  9     1     3  -0.0776 2.46e- 2
## 10     1     4  -0.115  3.84e- 2
## 11     1     5  -0.148  2.27e- 2
## 12     1     6  -0.159  4.78e- 4
```

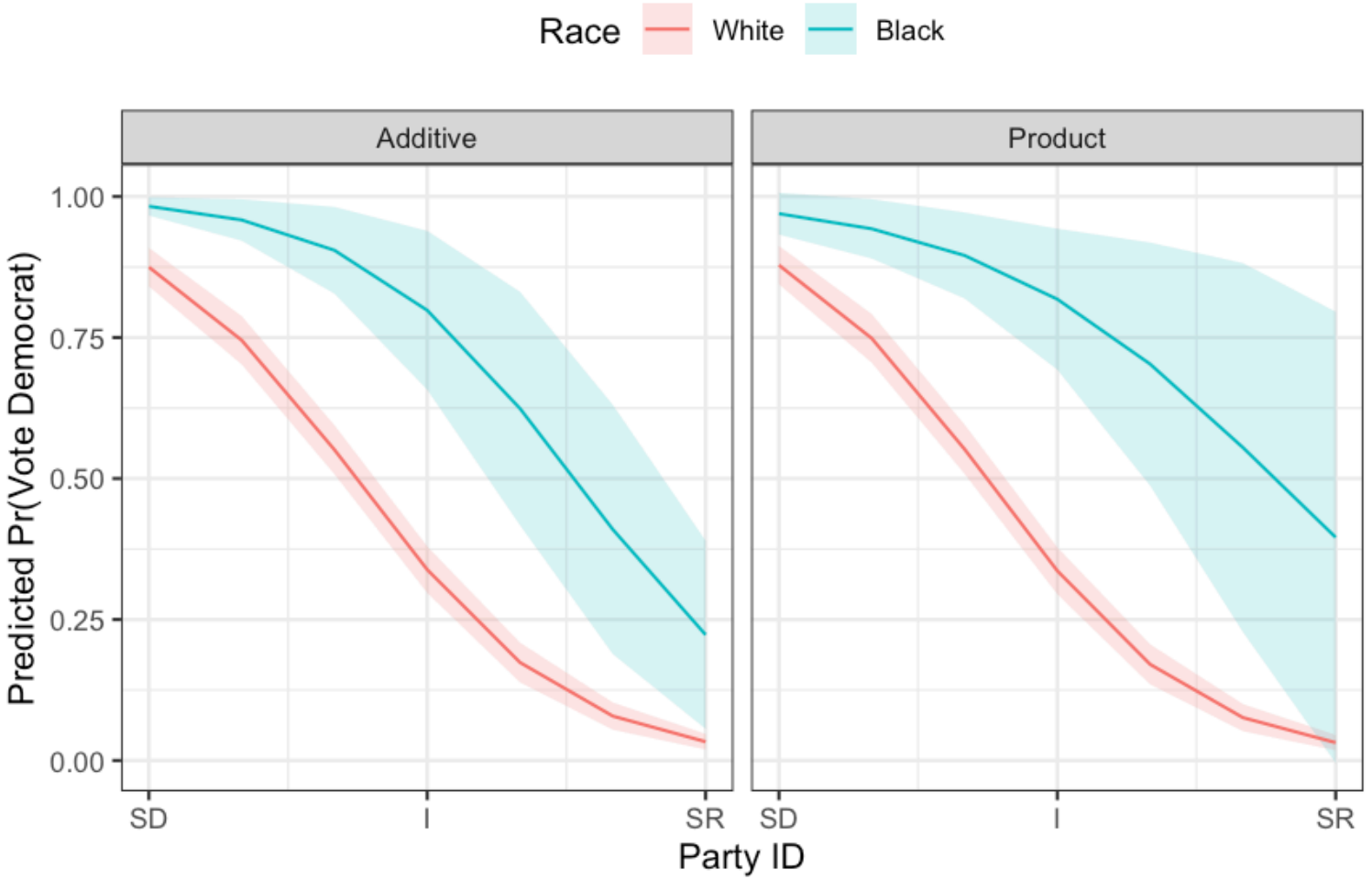
```
slp2a |>
  hypotheses(~pairwise | pid) |>
  as_tibble() |>
  left_join(
    anesu |> count(pid, black) |>
      pivot_wider(names_from = "black", values_from = "n", names_
  select(pid, estimate, p.value, starts_with("black"))
```

```
## # A tibble: 6 × 5
##     pid estimate  p.value black_0 black_1
##   <dbl>    <dbl>    <dbl>   <int>   <int>
## 1     1   0.103  1.01e-16     151      48
## 2     2   0.149  1.62e-12     142      24
## 3     3   0.138  1.71e- 4     140      11
## 4     4   0.0508 3.65e- 1      71       6
## 5     5  -0.0536 4.12e- 1     131       2
## 6     6  -0.115  1.25e- 2     161       2
```

```
lmtest::lrtest(add_mod, prod_mod)
```

```
## Likelihood ratio test
##
## Model 1: votedem ~ retnat + pid + age + male + educ + black + south
## Model 2: votedem ~ male + retnat + pid * black + age + educ + south
##   #Df  LogLik Df  Chisq Pr(>Chisq)
## 1   9 -401.05
## 2  10 -400.43  1 1.2288     0.2676
```

Effects:


Race
White
Black
Additive
Product
Predicted Pr(Vote Democrat)
1.00
0.75
0.50
0.25
0.00
SD
I
SR
Party ID

Example: ANES Voting Democrat

Two Numeric: `pid, age`

```
cmp_grid <- avg_comparisons(add_mod,
      newdata = datagrid(pid = 1:6,
                         age = quantile(anesu$age, c(.25, .75)),
                         grid_type = "counterfactual"),
      variables = list(pid = 1),
      by = c("pid", "age"))
cmp_grid |> as_tibble() |>
    select(pid, age, estimate, p.value)
```

```
## # A tibble: 12 × 4
##      pid   age estimate  p.value
##    <int> <dbl>    <dbl>    <dbl>
##  1     1    33  -0.132  6.32e-57
##  2     1    58  -0.109  4.53e-43
##  3     2    33  -0.190  6.03e-70
##  4     2    58  -0.173  1.74e-66
##  5     3    33  -0.200  3.73e-57
##  6     3    58  -0.204  2.69e-60
##  7     4    33  -0.154  1.93e-54
##  8     4    58  -0.175  6.96e-60
##  9     5    33  -0.0954 6.83e-46
## 10     5    58  -0.115  1.13e-51
## 11     6    33  -0.0516 6.01e-18
## 12     6    58  -0.0652 2.51e-24
```

```
cmp_grid |>
  hypotheses(~pairwise | pid) |>
  as_tibble() |>
  select(pid, estimate, p.value)
```

```
## # A tibble: 6 × 3
##     pid estimate p.value
##   <int>    <dbl>   <dbl>
## 1     1  0.0232   0.0216
## 2     2  0.0177   0.0275
## 3     3 -0.00413  0.181
## 4     4 -0.0205   0.0228
## 5     5 -0.0200   0.0222
## 6     6 -0.0136   0.0220
```

Example: ANES Voting Democrat

Two Numeric: `pid, age`

```
cmp_grid2 <- avg_comparisons(prod_mod,
      newdata = datagrid(pid = 1:6,
                         age = quantile(anesu$age, c(.25, .75)),
                         grid_type = "counterfactual"),
      variables = list(pid = 1),
      by = c("pid", "age"))
cmp_grid2 |> as_tibble() |>
    select(pid, age, estimate, p.value)
```

```
## # A tibble: 12 × 4
##      pid   age estimate  p.value
##    <int> <dbl>    <dbl>    <dbl>
##  1     1    33  -0.131  1.51e-64
##  2     1    58  -0.107  2.14e-35
##  3     2    33  -0.179  4.48e-43
##  4     2    58  -0.179  2.91e-55
##  5     3    33  -0.186  1.88e-32
##  6     3    58  -0.216  3.29e-45
##  7     4    33  -0.147  1.47e-41
##  8     4    58  -0.182  9.08e-51
##  9     5    33  -0.0958 7.71e-51
## 10     5    58  -0.115  5.15e-46
## 11     6    33  -0.0555 1.93e-20
## 12     6    58  -0.0621 8.52e-18
```

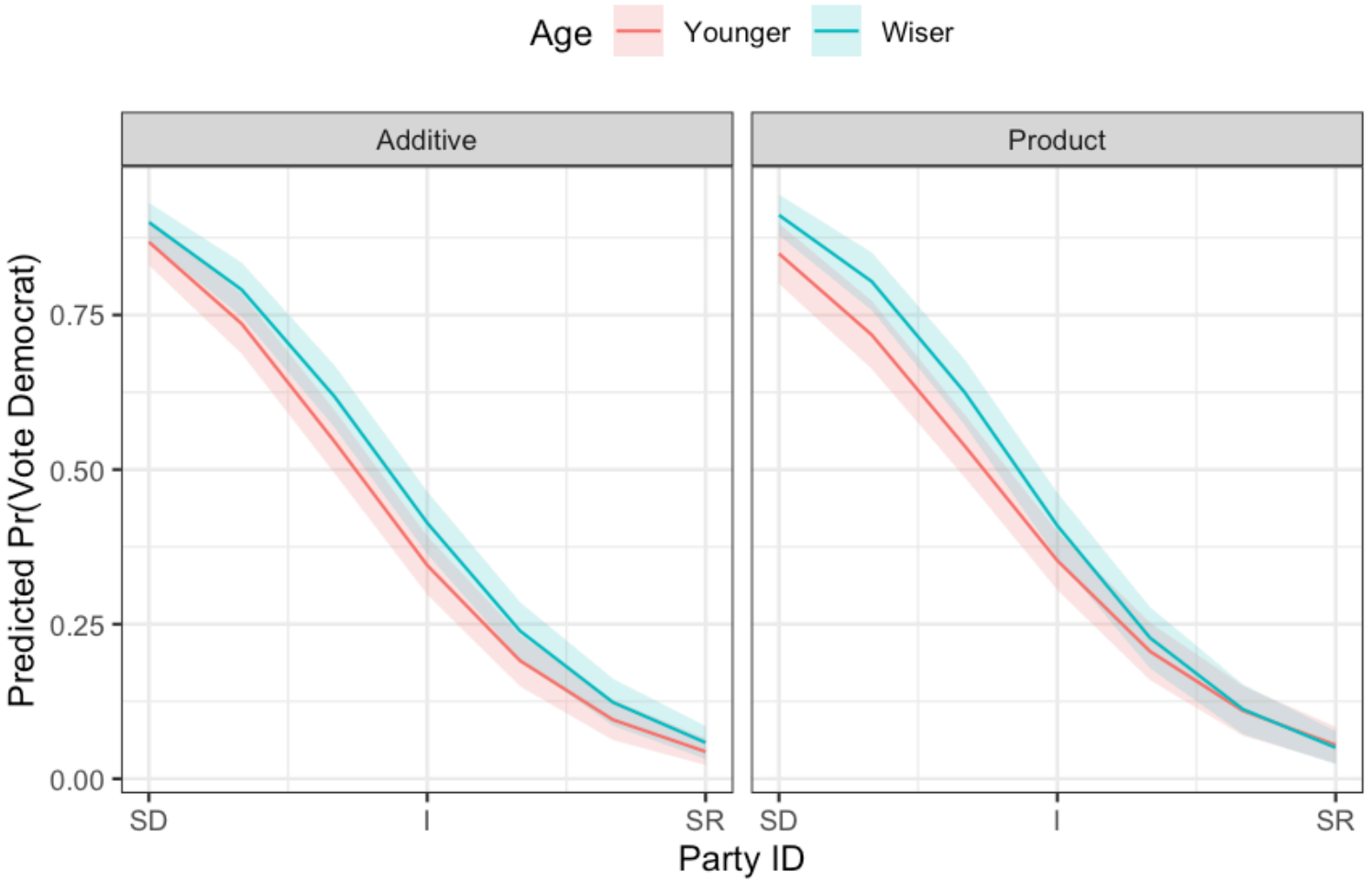
```
cmp_grid2 |>
  hypotheses(~pairwise | pid) |>
  as_tibble() |>
  select(pid, estimate, p.value)
```

```
## # A tibble: 6 × 3
##     pid  estimate p.value
##   <int>     <dbl>   <dbl>
## 1     1  0.0239   0.0228
## 2     2  0.000377 0.980
## 3     3 -0.0302   0.100
## 4     4 -0.0346   0.00954
## 5     5 -0.0196   0.0276
## 6     6 -0.00665  0.392
```

```
lmtest::lrtest(add_mod, prod_mod)
```

```
## Likelihood ratio test
##
## Model 1: votedem ~ retnat + pid + age + male + educ + black + south
## Model 2: votedem ~ male + retnat + pid * age + black + educ + south
##   #Df  LogLik Df  Chisq Pr(>Chisq)
## 1   9 -401.05
## 2  10 -399.98  1 2.1293     0.1445
```

Effects:


Age
Younger
Wiser
Additive
Product
Predicted Pr(Vote Democrat)
0.00
0.25
0.50
0.75
SD
I
SR
Party ID

Product Term can Reduce Moderation

```
g_add <- glm(y ~ x + z, data=ex_dat, family=binomial)
comps_add <- avg_comparisons(g_add,
                datagrid(z = c(0,1), grid_type = "counterfactual"
                variables = list(x = c(.25, .75)),
                by = "z")

comps_add
```

```
##
##  z Estimate Std. Error    z Pr(>|z|)    S  2.5 % 97.5 %
##  0   0.1685     0.0326 5.17   <0.001 22.0 0.1046 0.2323
##  1   0.0597     0.0131 4.54   <0.001 17.4 0.0339 0.0855
##
## Term: x
## Type: response
## Comparison: 0.75 - 0.25
```

```
comps_add |> hypotheses(~pairwise)
```

```
##
##   Hypothesis Estimate Std. Error     z Pr(>|z|)    S  2.5 % 97.5 %
##  (b2) - (b1)   -0.109     0.0229 -4.76   <0.001 19.0 -0.154 -0.064
```

```
g_prod <- glm(y ~ x * z, data=ex_dat, family=binomial)
comps_prod <- avg_comparisons(g_prod,
                datagrid(z = c(0,1), grid_type = "counterfactual"
                variables = list(x = c(.25, .75)),
                by = "z")

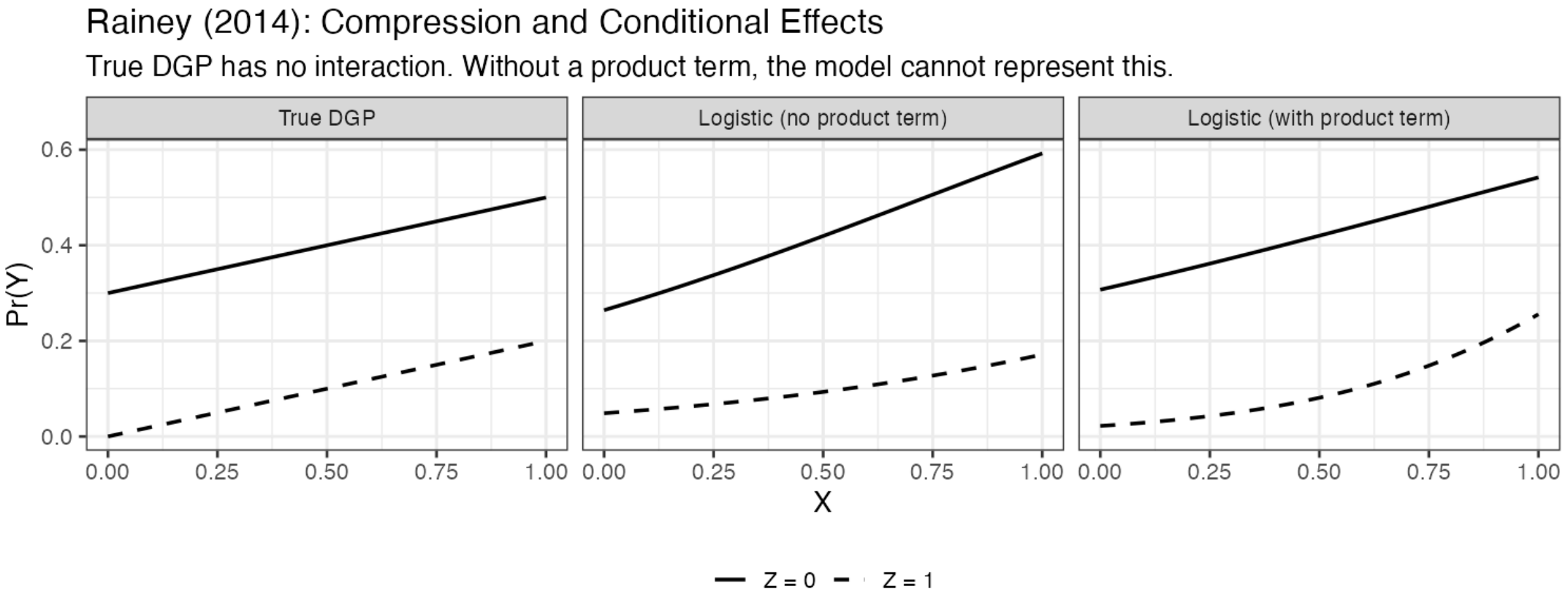
comps_prod
```

```
##
##  z Estimate Std. Error    z Pr(>|z|)    S  2.5 % 97.5 %
##  0    0.119     0.0377 3.16  0.00159  9.3 0.0451  0.193
##  1    0.105     0.0216 4.88  < 0.001 19.8 0.0631  0.148
##
## Term: x
## Type: response
## Comparison: 0.75 - 0.25
```

```
comps_prod |> hypotheses(~pairwise)
```

```
##
##   Hypothesis Estimate Std. Error      z Pr(>|z|)   S   2.5 % 97.5 %
##  (b2) - (b1)  -0.0135     0.0435 -0.312    0.755 0.4 -0.0987 0.0716
```

Reducing Moderation - What's Happening?



Effect Heterogeneity in GLMs.

In non-linear GLMs, effect heterogeneity is really about two questions (or two different sets of assumptions).

1. Is the functional form as estimated adequate (i.e., is a product term needed if not already included)?
2. Do the effects vary interestingly by subgroup?

Review

We covered the following topics:

1. Discuss Interaction Effects in the Linear Model
2. Discuss Interaction Effects in the GLM
3. Presentation of Interaction Results